PRUTHVIRAJ A RAI

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PROFESSIONAL SUMMARY

Computer Science undergraduate (B.E. 2024–2028, CGPA 8.0/10) at Mangalore Institute of Technology and Engineering (MITE) building practical systems that connect software, AI, and hardware. Hands-on project experience across IoT and embedded systems (ESP32, Telegram Bot API, web dashboards), computer vision (OpenCV, MediaPipe), and backend development (Node.js, Express.js). Selected for the second round of a national health-tech hackathon. Seeking an internship in IoT, embedded systems, or AI/computer vision.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, JavaScript, SQL

Embedded Systems and IoT: ESP32, Arduino IDE, Embedded C++, Tinkercad, sensor integration, Wi-Fi, HTTP/HTTPS, PySerial

Computer Vision and AI: OpenCV, MediaPipe, NumPy, SciPy, image processing, facial landmark tracking

Web and Backend: HTML, CSS, Node.js, Express.js, REST API design, Flutter (coursework)

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Figma, Canva, Microsoft Word, Microsoft Excel

Core Computer Science: Data Structures and Algorithms, Object-Oriented Programming, Databases, Operating Systems, Computer Networks

EDUCATION

2024 – 2028 (Expected)

Bachelor of Engineering (B.E.), Computer Science and Engineering — CGPA: 8.0/10

Mangalore Institute of Technology and Engineering (MITE), Karnataka, India

Relevant Coursework: Data Structures and Algorithms, Design and Analysis of Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Databases, Discrete Mathematics, Image Processing, Full Stack Web Development, Mobile Application Development (Flutter)

PROJECTS

Smart IoT Control System

ESP32, Embedded C++, Arduino, Telegram Bot API, HTML

https://github.com/Pruthvi-cs/Smart-IoT-Control-System-with-ESP32-and-TELEGRAM

- Built an ESP32-based IoT control system with two independent interfaces — a Telegram bot and a browser dashboard — for remote monitoring and control over Wi-Fi
- Implemented command handling, GPIO state management, and status reporting so both interfaces stay synchronized on the same device state
- Communicated with the board over HTTP/HTTPS and applied authorization and debugging practices for reliable remote operation

Driver Drowsiness Detection System (DDS)

Python, OpenCV, MediaPipe, NumPy, SciPy, PySerial, ESP32, L298N

https://github.com/Pruthvi-cs/Drowsiness_Detection_System-DDS

- Developed a real-time computer vision pipeline using MediaPipe facial landmarks and the Eye Aspect Ratio (EAR) to detect sustained eye closure from live webcam video
- Integrated Python-to-ESP32 serial communication (PySerial) to trigger alert hardware through an L298N motor driver
- Tuned detection thresholds to distinguish blinks from drowsiness events and reduce false alarms; documented as a working prototype, not an automotive-certified system

Women Safety Backend

Node.js, Express.js, JavaScript

https://github.com/Pruthvi-cs/women-safety-backend

- Designed a REST API for a women-safety IoT system with endpoints for creating emergency alerts, retrieving alerts, and reporting system statistics
- Structured the API as a clean integration surface for a frontend client or connected hardware device

Chrome Extensions

JavaScript, Chrome Extension APIs

https://github.com/Pruthvi-cs/Chrome_extensions

- Developed a Website Analyzer extension that inspects the active page and reports title, URL, H1/H2 headings, paragraph, image, link, and button counts, and internal vs external link distribution
- Built additional extensions including a to-do application, progressing through the Chrome extension permission and lifecycle model

EXPERIENCE AND ACHIEVEMENTS

Ayush Idea Hackathon 2026 — National Health-Tech Competition

- Selected for the second round with **AyuVari**, a project focused on digitizing and streamlining AYUSH healthcare services through technology

CERTIFICATIONS

- Deloitte Australia — Cyber Job Simulation
- OpenCV Bootcamp
- Google — Foundations of User Experience (UX) Design